

EXP 2 Analysis of SWR (Standing Wave Ratio) of Transmission Lines by Varying

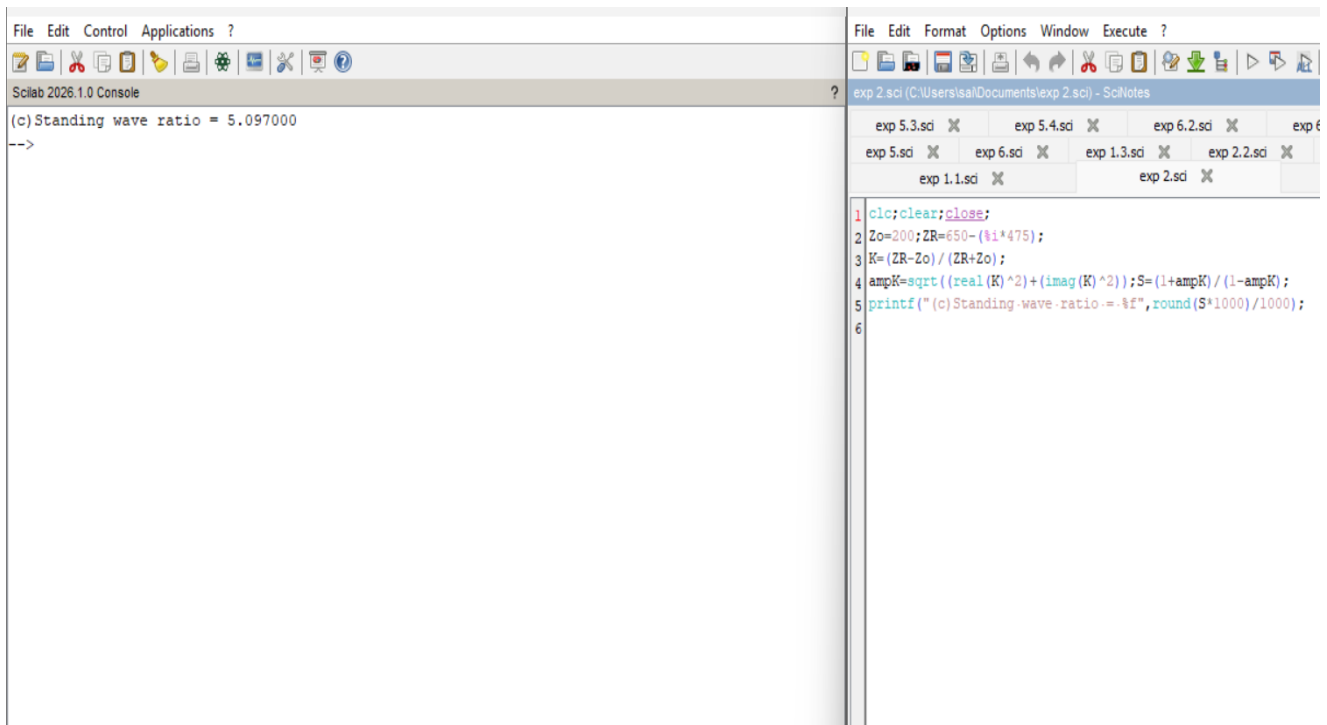
a) Load Impedance b) Characteristic Impedance

Test Case 1 — Varying Load Impedance (Z_L)

Analyze the SWR of a transmission line for a fixed characteristic impedance $Z_0 = 50 \Omega$, as the load impedance Z_L is varied from 650Ω to 850Ω in steps of 50Ω . When $Z_L = Z_0$, the SWR equals 1 (no reflection).

Scilab Program (update ZR per row and re-run)

```
clc; clear; close; Zo = 50; ZR = 650;
// set ZR per row K = (ZR-Zo)/(ZR+Zo);
ampK = sqrt((real(K)^2)+(imag(K)^2));
S = (1+ampK)/(1-ampK);
printf('(c)Standing wave ratio = %f', round(S*1000)/1000);
```



Result Table ($Z_L = 50 \Omega$, variable Z_o)

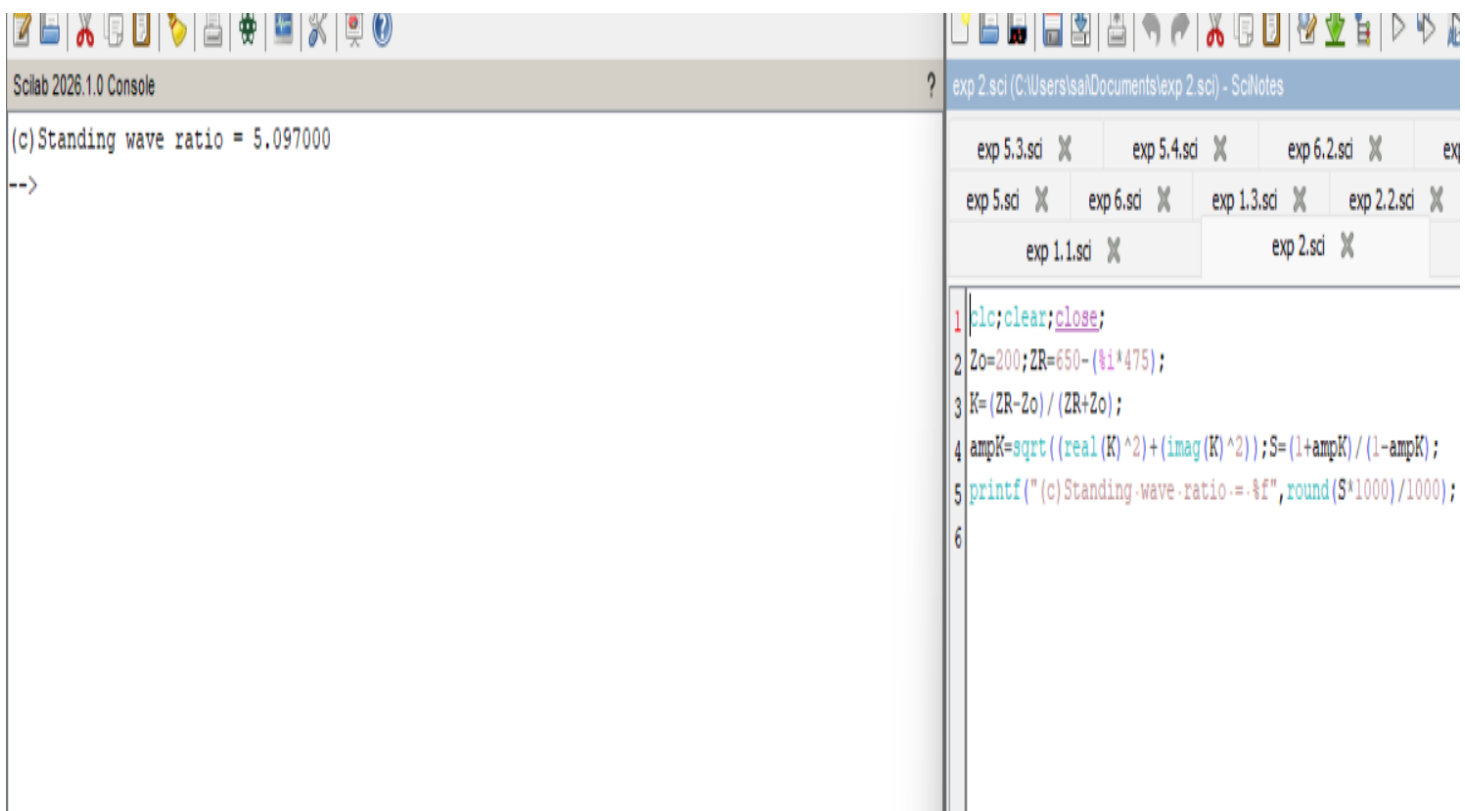
S.No	Z_o (ohms)	Simulation Results SWR	Theoretical Results SWR
1	650	13.000	13.000
2	700	14.000	14.000
3	750	15.000	15.000
4	800	16.000	16.000
5	850	17.000	17.000

Test Case 2 — Varying Characteristic Impedance (Z_o)

Analyze the SWR for a fixed load impedance $Z_L = 50 \angle 40^\circ \Omega$, while the characteristic impedance Z_o of the transmission line is varied from 250Ω to 400Ω in steps of 50Ω .

Scilab Program (update Z_o per row and re-run)

```
clc; clear; close; Zo = 250; ZR = 50*(cosd(40)+%i*sind(40));
// set Zo per row K = (ZR-Zo)/(ZR+Zo);
ampK = sqrt((real(K)^2)+(imag(K)^2));
S = (1+ampK)/(1-ampK);
printf('(c)Standing wave ratio = %f', round(S*1000)/1000);
```



Result Table (Z = 50∠40° Ω, variable Z)

S.No	Z _L (ohms)	Simulation Results SWR	Theoretical Results SWR
1	250	6.637	6.637
2	300	7.924	7.924
3	350	9.216	9.216
4	400	10.511	10.511

Result The SWR of the transmission line was analyzed for varying load impedance (Z fixed at 50 Ω) and for varying θ

characteristic impedance (Z fixed at 50∠40° Ω), and the simulation results (Scilab) matched the theoretical

values computed from $|K| = |(Z_L - Z_0)/(Z_L + Z_0)|$ and $SWR = (1 + |K|)/(1 - |K|)$.